

3-functor formalism

def.

$\mathcal{D} : \text{Core}(\mathcal{C}, \mathcal{E}) \rightarrow \text{Cat}_\infty$ is a lax symmetric monoidal functor.

$$- \otimes - := \mathcal{D} \left(\begin{array}{c} X \\ \swarrow \Delta \searrow \text{id}_X \\ X \times X \end{array} \right) : \mathcal{D}(X \times X) \rightarrow \mathcal{D}(X)$$

$\uparrow$ $\mathcal{D}$ being lax monoidal
 $\mathcal{D}(X) \times \mathcal{D}(X)$

and $1_{\mathcal{D}(X)} := \mathcal{D} \left(\begin{array}{c} X \\ \swarrow \text{id} \searrow \\ *_{\text{terminal}} \end{array} \right) : \mathcal{D}(*_{\text{terminal}}) \rightarrow \mathcal{D}(X)$
 $\uparrow$ $\mathcal{D}$ being lax monoidal
 $*_{\text{terminal}} \in \text{Cat}_\infty$

Notation $f: X \rightarrow Y$ in E

if $f_!$ admits a right adjoint denoted by $f^!$

dualizing object $\omega_f := f^!(1_{\mathcal{D}(Y)})$ in $\mathcal{D}(X)$.

unit object $1_{\mathcal{D}(X)} = f^*(1_{\mathcal{D}(Y)})$ by f^* being monoidal.

co-projection

$$f_!(f^!(-)) \otimes - \simeq f_!(f^!(-) \otimes f^*(-))$$

co-unital

where $\varepsilon: f_! f^! \rightarrow \text{id}_{\mathcal{D}(Y)}$

$$\begin{array}{c} \downarrow \varepsilon_- \otimes \text{id}_- \\ - \otimes - \end{array} \quad \searrow$$

induces (by adjunction)

$f: X \rightarrow Y$ in $\mathcal{E}$ is cohomologically smooth if,

Note: closed under composition
and arbitrary base change.
and contain all isomorphisms

1. $f_!$ admits a right adjoint $f^!$
2. ω_f is $\otimes$ -invertible in $\mathcal{D}(X)$
3. $\omega_{1_{\mathcal{D}(Y)}}, - : \omega_f \otimes f^*(-) \xrightarrow{\sim} f^!(-)$ is an isomorphism.
4. $\forall g: Y' \rightarrow Y$, f' satisfies 1., 2., 3.

and ω_f commutes with base change.

$$\begin{array}{ccc} X' := X \times Y' & \xrightarrow{f'} & Y' \\ g' \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

$$g^* f_! (\omega_f) \xrightarrow{\sim} f'_! g'^* (\omega_f)$$

Abstract Poincaré Duality

Def. category $LZ_{\mathcal{D}}^e$ of cohomological correspondences.

(as $(\infty, 2)$ -category)
Lu-Zheng

objects : objects of $\mathcal{C}$.

for any two objects X, Y

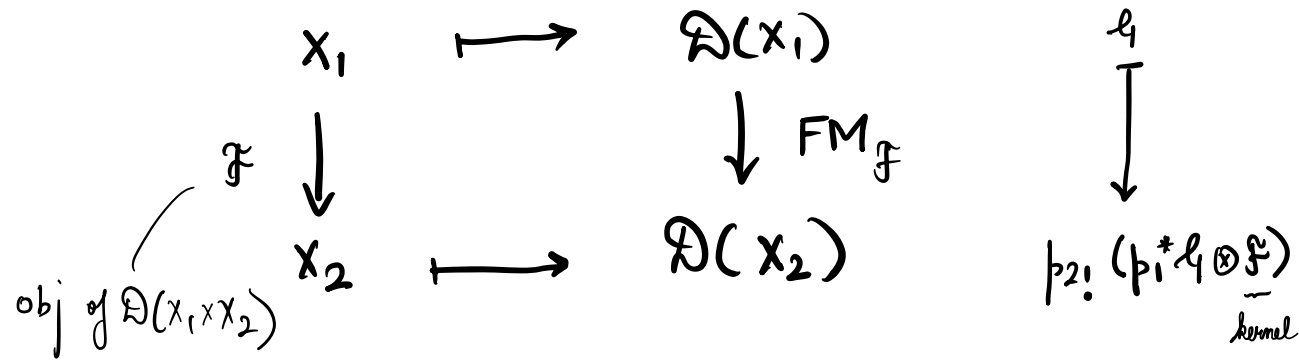
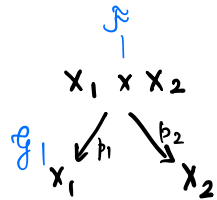
define $\text{Hom}_{LZ_{\mathcal{D}}^e}(X, Y) := \mathcal{D}(X \times Y)$

closed symmetric monoidal
 ∞ -category Cat_{∞} enrichment.

$$\begin{array}{ccccc} X_1 \times X_2 \times X_3 & & & & \\ p_{1,2} \swarrow & \downarrow p_{2,3} & \searrow p_{1,3} & & \\ X_1 \times X_2 & X_2 \times X_3 & X_1 \times X_3 & & \end{array}$$

[$\text{Corr}(\mathcal{C}, \mathcal{E})$

Def. Fourier-Mukai functor $FM : LZ_D^e \longrightarrow Cat_\infty$



functoriality

$$X_1 \xrightarrow{F_1} X_2 \xrightarrow{F_2} X_3$$

$$\begin{aligned} & FM_{F_2}(FM_{F_1}(\mathcal{L})) \\ &= q_{3!}(q_2^*(FM_{F_1}(\mathcal{L})) \otimes F$$

$$\begin{aligned}
 &= \kappa_{3!} \pi_{1,3!} \left(\pi_{1,3}^* \kappa_1^* \mathcal{L} \otimes \tilde{k} \right) \\
 &= \kappa_{3!} \left(\kappa_1^* \mathcal{L} \otimes \pi_{1,3!} \tilde{k} \right)
 \end{aligned}$$

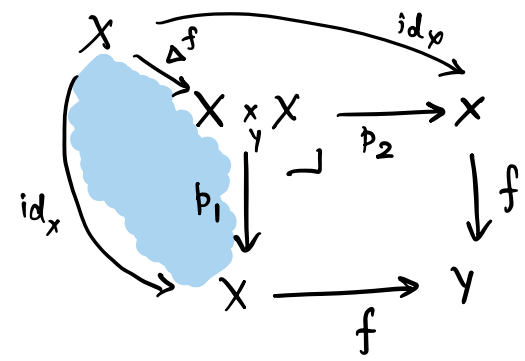
$$FM_{f_2 \circ f_1}(\mathcal{L}) = \kappa_{3!} \left(\kappa_1^* \mathcal{L} \otimes \pi_{1,3!} \underbrace{\left(\pi_{1,2}^* \mathcal{F}_1 \otimes \pi_{2,3}^* \mathcal{F}_2 \right)}_{\tilde{k}} \right)$$

10-min

Trace-cycle theory

$$f: X \rightarrow Y$$

and



A trace-cycle theory on f is a triple (L, tr, cl) of

1. a $\otimes$ -invertible object L in $\mathcal{D}(X)$
2. a morphism $tr: f_! L \rightarrow 1_{\mathcal{D}(Y)}$ in $\mathcal{D}(Y)$
3. a morphism $cl: \Delta_! 1_{\mathcal{D$

lemma 1.

Construction : $f: X \rightarrow Y$, $f \in E$, $g: Y' \rightarrow Y$

if f admits a trace-cycle theory (L, tr, cl)

then $(g'^* L, g^*(tr), \Delta^{f'}_! g'^* p_1! cl)$

is a trace-cycle theory on f' .

Proof: $g'^* L$ is $\otimes$ -invertible in $\mathcal{D}(X')$

$$g^* tr : g^* f_! L \xrightarrow{21} g^* 1_{\mathcal{D}(Y)}$$

$$\downarrow f'_! (g'^* L) \quad \downarrow 1_{\mathcal{D}(Y')}$$

$$\Delta^{f'}_! g'^* p_1! cl : \Delta^{f'}_! g'^* p_1! (\Delta^{f'}_! 1_{\mathcal{D}(X)}) \rightarrow \Delta^{f'}_! g'^$$

Square 1: claim

$$\begin{array}{ccccc}
 g'^* 1_{\mathcal{D}(x)} & \xrightarrow{\sim} & g'^* p_{1!}(\Delta^f_! 1_{\mathcal{D}(x)}) & \xrightarrow{g'^* p_{1!}(cl)} & g'^* p_{1!}(p_2^* L) \\
 \downarrow id_{\mathcal{D}(x)} & & & & \downarrow 1^2 \\
 g'^* 1_{\mathcal{D}(x)} & \xleftarrow{\sim} & \underbrace{g'^* f^*}_{\text{blue}} 1_{\mathcal{D}(y)} & \xleftarrow{g'^* f^*(tr)} & \underbrace{g'^* f^*}_{\text{green}} f_! L \\
 & & & & \underbrace{f'^* g'^* f_! L}_{f'^* f'_! g'^* L} \text{ base change } ///
 \end{array}$$

$$\begin{array}{ccccc}
 1_{\mathcal{D}(x')} & \xrightarrow{\sim} & \overbrace{p'_1!(\Delta^{f'}_! 1_{\mathcal{D}(x')})}^{id_{\mathcal{D}(x')}} & \xrightarrow{id_{\mathcal{D}(x')}} \underbrace{p'_1!(\Delta^{f'}_! g'^* p_{1!} cl)}_{\text{yellow}} & p'_1!(p_2'^* g'^* L) \\
 \downarrow id_{\mathcal{D}(x')} & & & & \downarrow 1^2 \\
 1_{\mathcal{D}(x')} & \x$$

lemma 2: Formal Poincaré Duality 1

let $f: X \rightarrow pt$ be in E .

let $A (= 1_{D(X)}) \in \text{Hom}_{LZ_{\mathcal{D}}^e}(X, pt)$

and $B \in \text{Hom}_{LZ_{\mathcal{D}}^e}(pt, X)$ be such that $A \dashv B$ in $LZ_{\mathcal{D}}^e$

($= I \in \text{obj } \mathcal{D}(X)$)

then $f_! : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$ admits a right adjoint $f^*(-) \otimes I : \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$.

Proof. Consider $FM : LZ_{\mathcal{D}}^e \rightarrow \text{Cat}_{\infty}$
 since it is a functor of $(\omega, 2)$ -categories, it preserves adjoints.

lemma 3 Formal Poincaré Duality II

let $f: X \rightarrow pt$ be in E . and it admits a trace-cycle theory (L, tr, cl)
 then $f_!$ admits a right adjoint given by $f^*() \otimes L$.
 [this will justify why trace-cycle was defined the way it was].

Proof. By lemma 2, it is enough to verify $A (= 1_{\mathcal{D}(X)}) \rightarrow B (= L)$
 in $LZ_{\mathcal{D}}$.

Step 1: finding co-unit

$$\varepsilon: A \circ B \rightarrow id_{pt}$$

ε is a 1-morphism
 in $\text{Hom}_{LZ_{\mathcal{D}}}(pt, pt)$
 $= \mathcal{D}(pt)$.

$$\begin{array}{ccc} f_!(id_X^* A \otimes id_X^* B) & \xrightarrow{\Delta_{pt}} & (1_{\mathcal{D}(pt)}) \\ \downarrow \script$$

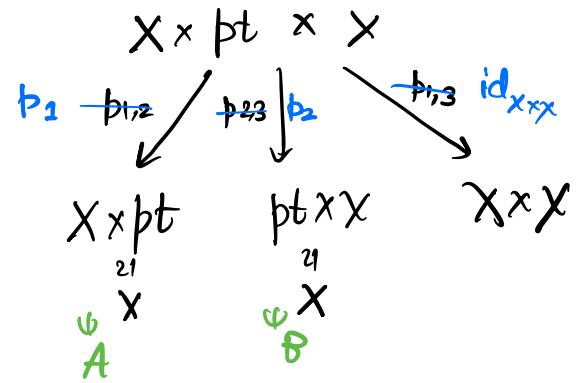
Step 2: Finding unit

η is a 1-morphism

in $\text{Hom}_{\mathcal{L}^{\text{re}}_{\mathcal{D}}}(X, X)$

$\mathcal{D}(X \times X)$

$$\eta: \underset{\substack{\uparrow \\ \Delta^*(1_{\mathcal{D}(X)})}}{\text{id}_X} \longrightarrow \underset{\substack{\uparrow \\ \text{id}_{X \times X}! (p_{1,2}^* A \otimes p_{2,3}^* B)}}{\underset{\substack{\uparrow \\ p_{2,3}^* B}}{\underset{\substack{\uparrow \\ p_2^* L}}{B \circ A}}}$$



Step 3.1 $A \circ \text{id}_X$

$$= p_{1!} (\text{id}_{X \times X}^* (\Delta_X! (1_{\mathcal{D}(X)})) \otimes p_2^* 1_{\mathcal{D}(X)})$$

$$\simeq p_{1!} (\Delta_X! (1_{\mathcal{D}(X)}) \otimes 1_{\mathcal{D}(X \times X)})$$

$$\simeq p_{1!} (\Delta_X! 1_{\mathcal{D}(X)})$$

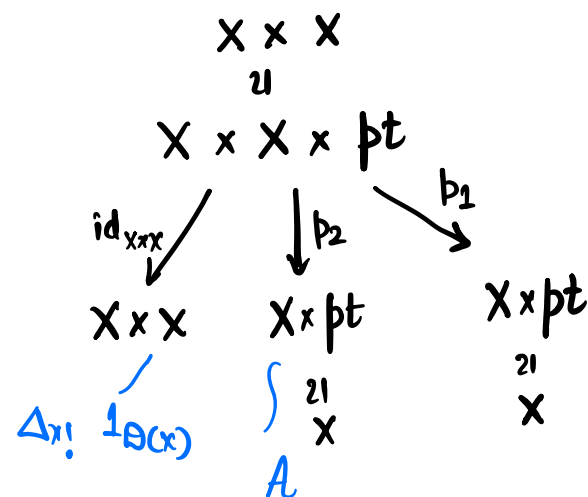
$$\simeq 1_{\mathcal{D}(X)}$$

$$= A$$

a.

$$1_{\mathcal{D}(X)}$$

$$\simeq p_{1!} (\Delta_X! (1_{\mathcal{D}(X)}))$$



$$b. \quad p_{1!}(\Delta_{X!}(1_{\mathcal{O}(X)})) \xrightarrow{p_{1!}(c)} p_{1!} p_2^* L$$

Can check $A \circ _$ is implemented by $p_1!$

and therefore map $b.$ is indeed $\text{id}_A \circ \eta$

Step 3.3 $(A \circ B) \circ A$

$$= \text{id}_{X!}(f^*(f_! L) \otimes \text{id}_X^* A)$$

$$= f^* f_! (L)$$

$$\begin{array}{ccccc}
 & & X & \times & \text{pt} & \times & \text{pt} \\
 & & \downarrow \text{id}_X & & \downarrow f & & \downarrow \text{id}_X \\
 A & \swarrow & X \times \text{pt} & & \text{pt} \times \text{pt} & & X \times \text{pt} \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & X & & \text{pt} & & X
 \end{array}$$

Step 3.4

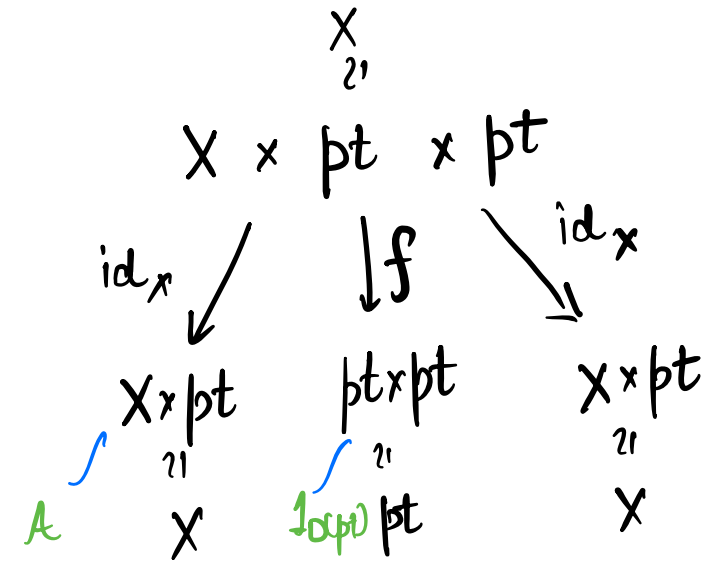
$$\text{id}_{\text{pt}} \circ A$$

$$1_{\mathcal{D}(\text{pt})} \quad 1_{\mathcal{D}(X)}$$

$$\approx \text{id}_X! \left(\text{id}_X^* (1_{\mathcal{D}(X)}) \otimes f^* (1_{\mathcal{D}(\text{pt})}) \right)$$

$$\approx 1_{\mathcal{D}(X)}$$

$$= A$$



$$d. \quad f^* f!$$

so we get

$$\begin{array}{ccccc}
 1_{\mathcal{D}(X)} & \xrightarrow{\sim} & p_{1!}(\Delta_{X!}(1_{\mathcal{D}(X)})) & \xrightarrow{p_!(cl)} & p_{1!}(p_2^* L) \\
 \downarrow id_{\mathcal{D}(X)} & & & & \downarrow \alpha \\
 1_{\mathcal{D}(X)} & \xleftarrow[\sim]{} & f^*(1_{\mathcal{D}(X)}) & \xleftarrow{f^*(tr)} & f^* f_!(L)
 \end{array}$$

Similarly for the other triangle

Therefore, if f admitting a trace cycle theory (L, π, cl)
 then $f_!$ admits a right adjoint $f^*(-) \otimes L$

Lemma 4 let $f: X \rightarrow Y$ be a morphism in E with a trace cycle theory (L, tr, cl) , then $f!$ admits a right adjoint given by $f^*(-) \otimes L$.

Proof: Define a category $\mathcal{C}_{/Y}^E$

object $f: X \rightarrow Y$ in E

morphism $g \in \text{Hom}_{\mathcal{C}_{/Y}^E}(f_1, f_2), g: X_1 \rightarrow X_2$ such that

$$\begin{array}{ccc} X_1 & \xrightarrow{g} & X_2 \\ f_1 \searrow & \circlearrowleft & \swarrow f_2 \\ & Y & \end{array}$$

commute.

(or give a filler)

$g \in E_{/Y}$ if $g \in E$ in $\mathcal{C}$.

terminal obj $id_Y: Y \rightarrow Y$ so

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ f \searrow &$$

and this induces a functor

$$i : \text{Corr}(\mathcal{C}_{/Y}^E, E_{/Y}) \longrightarrow \text{Corr}(\mathcal{C}, E)$$

This is a lax monoidal functor.

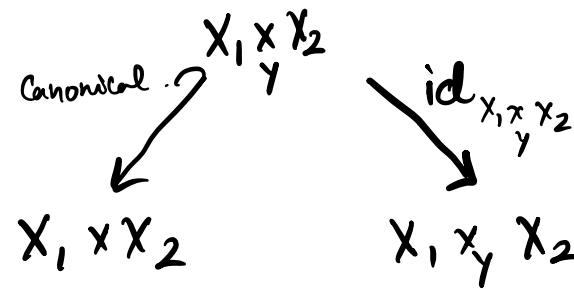
$$i(f) \times i(g) \longrightarrow i(f \times g)$$

$$X_1 \times X_2$$

$$X_1 \times_Y X_2$$

morphism

in $\text{Corr}(\mathcal{C}, E)$



1. an $\otimes$ -invertible object $\mathcal{D}_{/Y}(f) := \mathcal{D}(\text{dom } f) = \mathcal{D}(X)$
take L .

2. a morphism $f_! : L \rightarrow 1_{\mathcal{D}_{/Y}(\text{id}_Y)}$ in $\mathcal{D}_{/Y}(f)$. Note: since $\mathcal{Z}$ is identity on morphisms $f_! = f!$
 $1_{\mathcal{D}_{/Y}(\text{id}_Y)} \cong 1_{\mathcal{D}(Y)}$

take $\text{tr} : f_! L \rightarrow 1_{\mathcal{D}(Y)}$

3. a morphism $\Delta_!^f : \underbrace{1_{\mathcal{D}_{/Y}(f)}}_{1_{\mathcal{D}(X)}} \rightarrow p_2^*(L)$
in $\mathcal{D}_{/Y}(f \times f)$
 $\mathcal{D}(X \times_Y X)$

$$\Delta_!^f : \mathcal{D}_{/Y}(f) \rightarrow \mathcal{D}_{/Y}(f \times f)$$

$$\Delta_!^f : \mathcal{D}(X) \rightarrow \mathcal{D}(X \times_Y X)$$

by lemma 3 $\mathcal{D}(X) \xrightarrow{f_!} \mathcal{D}(Y)$
 $f_! : \mathcal{D}_Y(f) \rightarrow \mathcal{D}_Y(\text{id}_Y)$ admits a right adjoint

given by $f^*(-) \otimes L$.

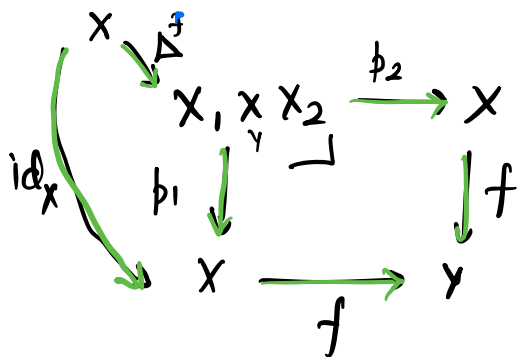
which is indeed $f^*(-) \otimes L$.

Corollary: If $f: X \rightarrow Y$ in E admits a trace cycle theory $(L, \text{tr}, \text{cl})$ then.
 $\forall g: Y' \rightarrow Y$, $f'_!$ admits a right adjoint $f^*(-) \otimes g'^*L$.

Theorem: $f: X \rightarrow Y$ in E is cohomologically smooth iff it admits a trace-cycle theory.

Proof: forwards implication
 given f is cohomologically smooth, the trace cycle theory is given by
 $(\omega_f, \alpha, \beta)$
 where $\alpha: f_! \omega_f \rightarrow 1_{\mathcal{D}(Y)}$ is the counit for the adjunction $f_! \dashv f^!$
 evaluated on $$

for β , consider



backward implication
 assuming f admits trace-cycle theory (L, tr, cl) then
 1. $f_!$ admits a right adjoint $f^*(-) \otimes L := f^!$ by definition

evaluate on $1_{D(Y)}$

$$f^*(1_{D(Y)}) \otimes L \cong 1_{D(X)} \otimes L \triangleq L = f^!(1_{D(Y)}) := \omega_f.$$

therefore $f^!$ is given by $f^*(-) \otimes \omega_f$

2. L is $\otimes$ -invertible and therefore ω_f is tensor invertible
 3. for $g: Y' \rightarrow Y$, $f^!$ has a right adjoint given by

$$f'^*(-) \otimes g'^* \omega_f = f'^!(-)$$

4. evaluate on $1_{D(Y')}$

Duality

$f: X \rightarrow Y$ in E , we have

$$\begin{array}{ccc} \mathcal{D}(X) & \begin{array}{c} \xrightarrow{f!} \\ \perp \\ \xleftarrow{f^*(\) \otimes \omega_f} \end{array} & \mathcal{D}(Y) \end{array} \quad \text{for } f \text{ cohomologically smooth}$$

this gives

$$\text{Hom}_{\mathcal{D}(Y)}(f! A, B) \cong \text{Hom}_{\mathcal{D}(X)}(A, f^*(B) \otimes \omega_f)$$

homotopy equivalence (ω -groupoids)

natural in $A \in \mathcal{D}(X)$
 $B \in \mathcal{D}(Y)$

let us put $B = 1_{\mathcal{D}(Y)}$

$$\text{Hom}_{\mathcal{D}(Y)}(f! A, 1_{\mathcal{D}(Y)}) \cong \text{Hom}_{\mathcal{D}(X)}(A, \omega_f)$$

Let $\mathcal{C} =$ Locally compact Hausdorff spaces and E be all morph

$$\cong R\Gamma(X, (A \otimes \omega_f^{-1})^\vee)$$

$$\cong R\Gamma(X, \omega_f^{-1\vee} \otimes A^\vee) \cong R\Gamma(X, A^\vee \otimes \omega_f)$$

LHS take $Y = \text{pt.}$

$$\underbrace{\text{Hom}_{\text{DAb}}(R\Gamma_c(X, A), \mathbb{Z}[0])}_{\text{is already internal hom}} = R\Gamma_c(X, A)^\vee$$

Duality $\checkmark$

$$R\Gamma_c(X, A)^\vee \cong R\Gamma(X, A^\vee \otimes \omega_f)$$

let $A = \mathbb{Z}_x[0]$ for cohomology. $f: X \rightarrow \text{pt}$

$$A^\vee = A$$

$$\boxed{R\Gamma_c(X, \mathbb{Z}_x[0])^\vee \cong R\Gamma(X, \omega_x)}$$

Remarks: 1) $f: \mathbb{R} \rightarrow *$ is cohomologically smooth
[give a trace-cycle theory]

for $\omega_f = \mathbb{Z}_{\mathbb{R}}[1]$

2.) All manifold bundles are cohomologically smooth.

3.) for $f: X \rightarrow *$ where X is oriented d -dim manifold
 $\omega_X \simeq \mathbb{Z}_X[d]$ dualizing object.

4.) A six functor formalism $\mathcal{D}: \text{Corr}(\mathcal{C}, \mathcal{E}) \rightarrow \text{Cat}$ is
geometric if any smooth morphism in $\mathcal{C}$ is cohomologically smooth w.r.t $\mathcal{D}$.

Reference: Poincaré Duality Revisited

Bogdan Zavyalov 2023

Six-Functor Formalism

Peter Scholze 2022/23